

Plug Puller — Starter Guide

From “that plug is stuck” to a printed tool: figure out your plug (match it or measure it), fill in the Customizer steps, print.

The last page of this PDF is a 1:1 measuring stencil sheet. Print this document at 100% scale / “Actual size” — never “fit to page” — and check the 50 × 50 mm calibration square on that page before trusting it.

What you need

The plug you want to pull (in its outlet), and either the printed **measuring stencil** (stl/Measuring_Stencil.stl, 1.2 mm cards, no supports) or the paper stencil sheet at the back of this document.

The stencil cards — legend

Each 3D-printed card has a raised two-letter ID you can find by touch:

P1	plug silhouette: flat 2-prong lamp plug (NEMA 1-15)
P2	plug silhouette: standard 3-prong plug (NEMA 5-15)
P3	plug silhouette: heavy-duty extension cord (NEMA 5-15)
R1	ruler — raised mm ticks, numerals every 10 mm, tactile edge notches
C1	cord gauge — through-holes Ø 3–9 mm
F1 / F2	finger sizing — 18 labeled holes (Ø 15–25 on F1, Ø 26–32 on F2)

Each P card has three openings: **W** (the plug's width outline), **T** (its thickness outline), and a round cord hole. It is a “does my plug match this preset?” test — no numbers involved.

Path A — match a preset (fastest, no numbers)

1. Take the **P1**, **P2**, and **P3** cards to your plug.
2. Hold each card's **W** cutout over the plug (wide side), then the **T** cutout (thin side). Try the cord in the round hole.
3. If the plug fills one card's openings — snug, no big gaps — that preset is your plug. Pick it in Step 1:
 - P1 → Flat 2-prong lamp plug - NEMA 1-15
 - P2 → Standard 3-prong plug - NEMA 5-15
 - P3 → Heavy-duty extension cord - NEMA 5-15
4. Skip ahead to “Fill in the Customizer steps” on the next page.

No card fits? Your plug is between presets — take Path B; the measured tool will fit better than any preset anyway.

Path B — measure (about 5 minutes)

Work the worksheet in docs/guides/measuring-guide.md; the numbers below are the worksheet's own numbering.

1	Plug length — R1 , wall plate to the plug's back face
2–3	Plug width near wall / near cord — R1 notched edge against the plug body
4–5	Plug thickness near wall / near cord — R1 , across the thin direction
6	Cord thickness — C1 , smallest hole the cord slides through
7	Wall plate style — a picture quiz, see the measuring guide
8	Finger knuckle width — F1 / F2 , smallest comfortable hole, minus 5
9	Hand width — R1 , across the four knuckles, flat hand

openscad-plug-puller · starter guide · page 1 of 3

8 and 9 are only needed if you pick “Measure my hand” in Step 2 — the built-in Small / Medium / Large cover most hands.

Fill in the Customizer steps

Open `src/Plug_Puller_Parametric.scad` in OpenSCAD and show the Customizer panel (uncheck *View ▸ Hide Customizer*). Every click is spelled out in `docs/guides/quick-start-beginner.md`; the short version:

Step 0 — Tool Style	leave on <code>Auto</code> from <code>plug</code> . Thick round plugs (like P3) get the heavy-duty clamshell, everything else the flat tool.
Step 1 — Your Plug	the preset from Path A, or your Path B numbers with <code>plug_preset = Measure my plug</code> .
Step 2 — Size	Medium fits most adults; or <code>Measure my hand</code> with worksheet numbers 8–9.
Step 3 — Attachment	how the tool attaches to the plug. Keep the default <code>Zip ties + Velcro</code> unless you know you want less.
Step 4 — Cord Hook	Right or Left (flat tool only).

Press **F6** to render, then *File ▸ Export ▸ STL*. The clamshell is one plate — print it **twice**, flip one copy, zip-tie the pair face to face around the plug.

Which steps shape which tool

Each step says which tool it shapes; settings for the other tool are simply ignored (the console tells you which ones).

Step	Flat tool	Heavy-duty clamshell
0 — Tool style	picks it	picks it
1 — Your plug	pocket, notch, hook slot	arm gap, arm length, cord channel
2 — Size	finger holes, body size	finger bores
3 — Attachment	zip-hole grid, velcro wings	zip stations, arm strap slots
4 — Cord hook	J-hook direction	<i>ignored — no cord hook</i>

On the clamshell, the Step 3 **zip ties are what hold the two plates together** — leave them on unless you have another plan (the model warns you if you turn them off).

Printing the 3D stencil on a small bed

The cards pack onto print sheets automatically. Open `Measuring_Stencil.scad`, set `bed_width` and `bed_depth` to your printer's bed, and the console lists which cards land on which sheet. Render with `part_index = 1`, export and print, then repeat with `part_index = 2`, and so on. (`part_index = 0` previews all sheets at once — don't print that one.)

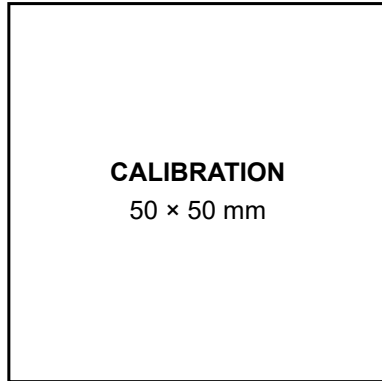
If the print doesn't fit

One symptom, one number to nudge: `docs/guides/fit-troubleshooting.md`. Want to sanity-check before printing the real tool? Print a paper 1:1 outline sheet of your combination first (`docs/guides/print-preview-outlines.md`).

Next page: the 1:1 measuring stencil sheet. Cut along the dashed lines. Measure the calibration square before trusting anything — it must be exactly 50 × 50 mm.

Plug Puller — Measuring Stencil Sheet

Print at 100% scale / “Actual size” — never “fit to page”. Check the calibration square before trusting anything on this sheet.

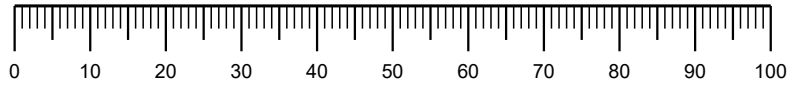


1. Measure this square with a ruler.
It must be exactly 50 × 50 mm — if not, your print was scaled: re-print at 100%.

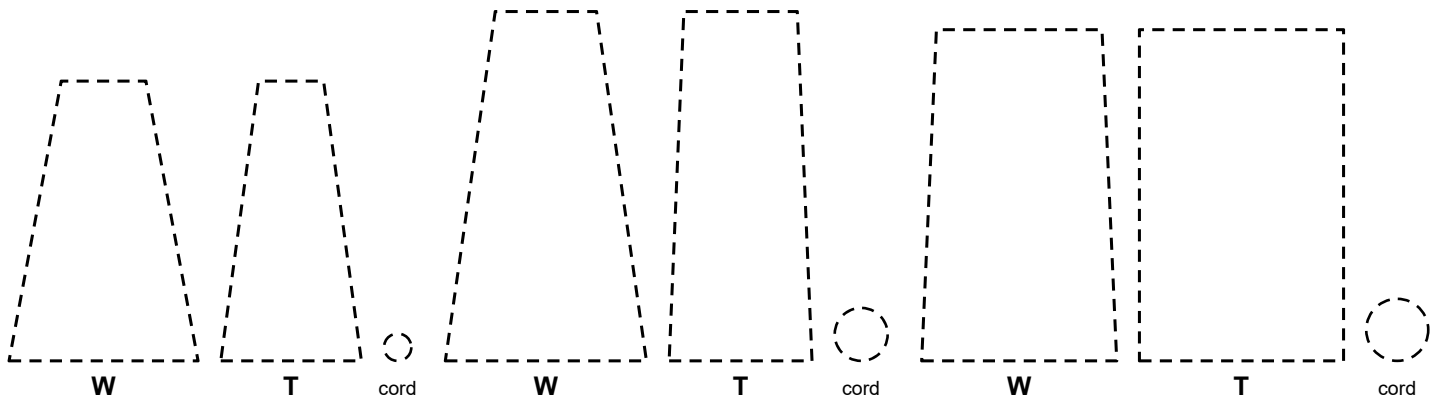
Card legend: <docs/guides/starter-guide.md>

Works on A4 and US Letter paper.

R1 — ruler (mm): plug length, widths, thicknesses, hand width



2. Plug silhouettes (1:1) — hold your plug against **W** (wide side) and **T** (thin side); if it matches, pick that preset in Step 1.



P1 Lamp 2-prong NEMA 1-15

P2 Standard 3-prong NEMA 5-15

P3 Heavy-duty cord NEMA 5-15

3. **F1 / F2 — finger circles: smallest circle your middle finger passes through, minus 5 = finger width.**

